

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

May 2013

IT402: Wireless communication and Mobile Computing

Date: 08.05.2013, Wednesday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Write the answer of following Questions. [11]

(a) Explain following terms [2]

(i) FDMA

(ii) TDMA

(iii) CDMA

(iv) MAC

(b) What are the features of CDMA over GSM network? [2]

(c) How does registration carried out in Mobile IP? [2]

(d) What is Inter Frame Spacing with respect to Wireless LAN? What is the time duration of each interface spacing? [2]

(e) What is the basic purpose of DHCP? Name the entities of DHCP. [3]

Q - 2 Write the answer of following Questions. [12]

(a) What is Multiple Access Procedure? Why it is important? Describe FDMA, TDMA, CDMA and SDMA with application areas and examples. [5]

(b) Draw and explain Protocol Architecture of 802.11 [7]

OR

(b) Draw and explain Bluetooth Protocol Stack. [7]

Q - 3 Write the answer of following Questions. (Attempt any TWO) [12]

(a) Describe security and authentication services offered by GSM.

(b) What are the entities and terminologies of Mobile IP? What are the requirements for development of Mobile IP?

(c) What are the advantages of Wireless LAN?

SECTION – II

Q - 4 Write the answer of following Questions. [11]

(a) List the disadvantages of In-direct TCP. [2]

(b) What is role of The Supplicant in 802.1x Authentication? [2]

(c) What is Piconet and Scatternet? [2]

(d) What is Transaction-oriented TCP? [2]

(e) Write down the difference between Infrastructure and Adhoc network. [3]

Q - 5 Write the answer of following Questions. [12]

(a) What are the applications of Modulation and Multiplexing in wireless communication? Can wireless communication work without Multiplexing and Modulation? [5]

(b) Draw the block diagram of GSM architecture and explain all entities in brief. [7]

OR

(b) Draw the block diagram of GPRS architecture and explain all entities in brief. [7]

Q - 6 Write the answer of following Questions. (Attempt any TWO) [12]

(a) What is WEP? Why is it considered unsafe? What are the mechanisms advised to ensure security in Wireless LAN?

(b) Write a short note on Satellite Network.

(c) HiperLAN2 networks can operate in two different modes. Explain those two modes.